THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL II EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT (CIVL2008 / CIVL2108)

May 13, 2013 9:30 am – 12:30 pm (3 hours)

Answer **ANY FOUR** questions

All questions carry equal marks

Use of Calculators

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

QUESTIONS

1. After the opening of the new campus, the University would like to initiate a series of renovation works to convert the vacated office spaces located in various existing academic building towers into useful teaching and learning facilities. To avoid causing any unnecessary nuisance, these renovation works will be carried out progressively over a rather long time period.

   a) In view of the nature of work, explain to the senior management of the university what are a lump sum contract and a remeasurement contract and distinguish their advantages and disadvantages when applied to the renovation works mentioned above. [17 marks]

   b) The university would like to seek your advice on the principle and operation of a term contract. To facilitate the Senior Management to make a final decision, you are requested to provide the detailed reasons justify whether this contractual arrangement is the most appropriate approach for this project. [8 marks]

2. The following table shows the duration and logical relationship of the activities involved in a construction project:

Activity ID	*Predecessor*	*Duration (month)*
A	--	1
B	--	3
C	--	4
E	B	4
F	B	3
G	C, F	1
H	B	1.5
I	H, E	2.5
J	H, E	2
K	C, F, J, M	2
L	K	4.5
M	A, B	1

a) Describe the concept of an "activity-on-arrow" network and highlight the major disadvantage of this method. [3 marks]

b) Make use of the incomplete "activity-on-node" network diagram in Appendix I (*note:* by detaching Appendix I from the question paper and attaching it to your answer book) to show the logical sequence of the project and the starting and finishing dates of the activities. [9 marks]

c) Prepare a table showing the early and late start and completion dates and work out the total and free floats available for each activity. [8 marks]

d) Describe what measures should be taken by the project manager to monitor and control the project progress. [5 marks]

3. Seizing the opportunities brought by the ten mega infrastructure projects and the increasing demand for residential properties, a small contractor is prepared to expand its company size to target a series of medium scale construction projects. Being a family business, the company is rather unorganised with its employees working on various tasks as assigned by the Managing Director. Besides, the Managing Director likes to coordinate and oversee the construction projects himself.

a) While more projects are anticipated after the expansion, a clearer line of authority and responsibility would be inevitable. Identify at least five functional units the contractor should establish and prepare an organisational chart accordingly. Describe the theory behind functional organisation and outline the pros and cons of this organisational type. [10 marks]

b) To improve the performance and efficiency of the prospective projects, advise the company whether they should consider employing a project-based or matrix organisational structure? Support your argument with the aid of suitable diagrams and detailed justifications. [15 marks]

4. a) *"The calculation of maximum land price using the residual method is inaccurate, subjective and unreliable"*. Comment on this statement by making reference to the major components and assumptions one would normally consider to arrive at the land price. [18 marks]

   b) Describe the major risks that could affect the accuracy of the land price calculation. Explain whether it is possible to analyse the effect of risks through the residual method. [7 marks]

5. a) Compute the unit rate for a 150mm thick in-situ reinforced concrete (1:2:4, 40mm aggregate) sloping roof slab sized 20m × 10m based on the basic cost data as shown below:

Material

Type	*Material price*
Cement	$900.00 per tonne
Sand	$130.00 per tonne
Aggregate	$110.00 per tonne

Labour

Type	*Wage*
Concrete vibration	$140.00 per hour
All other workers	$120.00 per hour

Location	*Labour Constants*	
	Man hour	*Gang hour*
Slab < 100mm thick	5.32	1.33
Slab 100-150mm thick	4.68	1.17
Slab 150-300mm tick	4.32	1.08

Plant

Type	*Hiring charge*
Mixer (output = 0.333 hour / m^3)	$180.00 per hour
Pump (output = 0.25 hour / m^3)	$240.00 per hour

Note: Previous experience indicates that a sloping slab is 1.15 times more difficult to cast than a typical floor slab, and a different member size could increase the time by up to a 13.7%. [20 marks]

   b) The contactor is considering using a pricing strategy when pricing the tender. Explain with the assistance of a suitable example what is *"front loading"* and *"rear loading"*, and how should the client deal with the front or rear loaded tenders. [5 marks]

~ END OF PAPER ~

Appendix I